

Uniform vs. heterogeneous alphabets

Breaking Assumption 1(iii) at matched total rate

DDMS simulator · Gaussian shift, $p = 1/2$ · exact error probability, no Monte Carlo

Setup

Two schemes spend an identical communication budget of R bits. VANILLA is the article's setting: every sensor uses the same $M = 4$ quantizer, 2 bits each, so $|U^i|$ is uniform and Assumption 1(iii) holds. MMIX gives a fraction c of the sensors a 1-bit $M = 2$ quantizer and the rest the $M = 4$ one, so alphabets differ across sensors and the assumption fails. Thresholds are nested, $\{1\} \subset \{0.2, 1, 5\}$, so alphabet size is the only variable.

Matching the budget fixes the sensor count: $N = R/(2 - c)$. Spending bits on coarser sensors therefore buys *more* of them — at $R = 240$, VANILLA fields 120 sensors and all-1-bit fields 240. The metric is the total error exponent $-\log J^N$; each 2.3 units is one more decimal place of reliability.

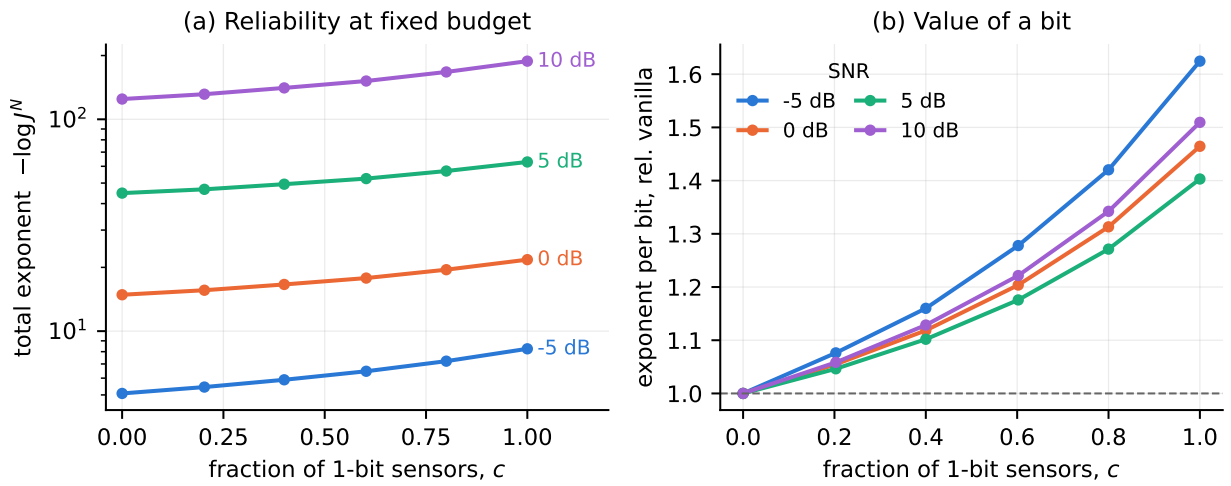


Figure 1: At $R = 240$ bits. (a) Exponent rises monotonically as budget shifts to 1-bit sensors, at every SNR. (b) The same effect per bit: a bit is worth 1.4–1.6 $\times$ more when spent on a new sensor than on finer resolution.

Results at $R = 240$ bits

	SNR			
	-5 dB	0 dB	5 dB	10 dB
VANILLA exponent ($c = 0$, $N = 120$)	5.08	14.85	44.83	124.64
MMIX exponent ($c = 0.6$, $N = 171$)	6.47	17.80	52.49	151.62
All 1-bit ($c = 1$, $N = 240$)	8.25	21.75	62.90	188.14
Error reduction, $c = 0 \rightarrow 1$	$10^{1.4} \times$	$10^{3.0} \times$	$10^{7.8} \times$	$10^{27.6} \times$
Mixing penalty (exact/average)	0.997–1.024 over all 48 mixed banks			

VANILLA — uniform $|U| = 4$

Pros. Inside Assumption 1(iii), so the article's exponent results apply directly. Exchangeable sensors give the cheapest exact computation (one multinomial). Fewest sensors for a given budget — the right choice when hardware, not bandwidth, is the binding constraint.

Cons. Weakest configuration per bit at every SNR tested. The second bit only refines confidence, while a first bit buys an independent opinion; at $R = 240$, 0 dB it gives up a factor 10^3 in error probability against the same budget spent on 1-bit sensors.

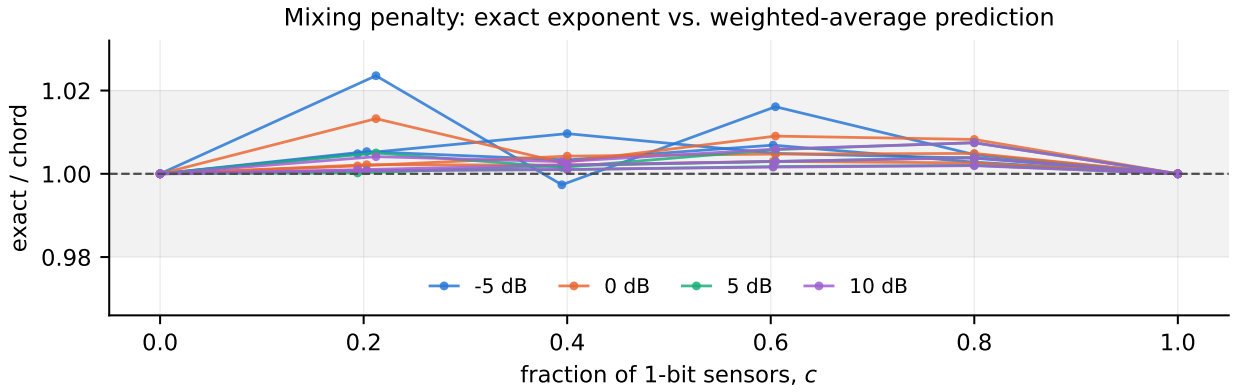


Figure 2: Every mixed bank, all four SNRs and all three budgets (60/120/240 bits), against the weighted average of the two uniform endpoints. Deviations stay inside $\pm 2\%$ and shrink as the budget grows: heterogeneity itself is free.

MMIX — heterogeneous $|U| \in \{2, 4\}$

Pros. Strictly better exponent than VANILLA at equal budget, tunable continuously through c . Costs nothing beyond the reallocation itself — performance is the rate-weighted average of the uniform schemes to within 2%. Practical when sensors are genuinely unequal (different hardware, links, or battery), since a mixed fleet loses no efficiency.

Cons. Outside Assumption 1(iii), so the article’s analysis does not formally cover it. Needs more sensors and a fusion center that tracks per-sensor γ^i . Exact evaluation is costlier (cross-group convolution). And the gain is not from heterogeneity: the uniform all-1-bit bank is better still.

Conclusions

- Assumption 1(iii) is a proof convenience, not a performance boundary.** Mixed banks land on the straight line between uniform ones. Given the hypothesis, sensors are independent and their log-likelihood contributions simply add — the fusion center is indifferent to whether they came from matching alphabets.
- The real lever is rate allocation, not uniformity.** Under a bit budget, more coarse sensors beat fewer fine ones at every SNR: 1-bit sensors convert bits into exponent $1.4\text{--}1.6\times$ more efficiently. The effect strengthens with SNR, where a single threshold already separates the hypotheses well.
- Read exponents as digits of reliability.** Because errors decay exponentially, the modest gap $14.85 \rightarrow 21.75$ at 0 dB is three orders of magnitude in J^N .

Caveat: measured at fixed nested thresholds $\{0.2, 1, 5\}$. Whether $M = 4$ with Chernoff-optimized thresholds closes the per-bit gap is untested. Static heterogeneity only — alphabets are fixed in advance, not adapted to observations. Source: runs/Mmix/csweep_M2M4_*, 2026-08-03.